

# Yaoxuan (Thaddy) Wu

✉ thaddywu@gmail.com • 🌐 thaddywu.github.io

## Education

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**University of California, Los Angeles**  
*Ph.D. Candidate, Department of Computer Science*  
**Expected Graduation Date: 06/2027**

**Los Angeles, CA**  
*Sep. 2022 – Present*

**University of California, Los Angeles**  
*M.S., Department of Computer Science*

**Los Angeles, CA**  
*Sep. 2022 – Mar. 2025*

**Peking University**  
*B.S., Summa Cum Laude, EECS*  
*Turing Class (most selective undergraduate program in China)*

**Beijing, China**  
*Sep. 2018 – Jun. 2022*

## Internship

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**Amazon Web Services**  
*Research Scientist Intern*

**Arlington, VA**  
*Jun. 2025 – Sep. 2025*

## Research Interest

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Program Testing and Verification, Database, Big Data Analytics, LLM4SE

## Project Experience

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**AWS Cloud Ops**  
Intern Applied Scientist  
Mentor: **Dr. Brandon Paulsen**, Manager: **Dr. Joey Dodds**

**AWS**  
*Jun. 2025 - Sep. 2025*

- **CloudGym: A High-Coverage Cloud Emulator**

[1] In preparation.

- A large-scale LLM-based cloud emulator that enables AWS users to execute and test cloud applications safely in a local sandboxed environment.
- I design three techniques to improve the scalability of cloud emulation and the accuracy of emulated API responses, achieving 69% correctness compared to LocalStack's 38%.
- CloudGym can successfully emulate Terraform projects that create up to 40 AWS resources and issue over 250 API calls, demonstrating both scalability and high coverage.

**UCLA SEAL**  
Graduate Student Researcher  
Advisor: **Prof. Miryung Kim**

**UCLA**  
*Sep. 2022 - Present*

- **PALM: Path-aware LLM-based Test Generation**

[2] Under submission.

- An LLM-based test generator targeting for specific program paths.
- We leverage LLM to understand the input-output relationships introduced by API calls, which are challenging for traditional symbolic executors.
- Our tool achieves  $1.41\times$  higher path coverage than Java Pathfinder, and  $1.31\times$  higher path coverage than direct LLM generation.

- **NaturalSym: Natural Symbolic Execution-based Testing for Big Data Analytics**

[3] **Yaoxuan Wu**, Ahmad Humayun, Muhammad Ali Gulzar, and Miryung Kim. Natural Symbolic Execution-based Testing for Big Data Analytics. *FSE 2024*.

- A novel symbolic execution-based test generation tool for scala-based Apache Spark applications.
- We extend existing SMT solvers to satisfy both path conditions from symbolic execution and input semantics from user specification.
- Our tool is the first to incorporate input naturalness in constrained-based testing that generates both high path-coverage and natural tabular test inputs.

- **NaturalFuzz: Natural Input Generation for Big Data Analytics**

[4] Ahmad Humayun, **Yaoxuan Wu**, Miryung Kim, and Muhammad Ali Gulzar. NaturalFuzz: Natural Input Generation for Big Data Analytics. *ASE 2023*.

- A mutation-based fuzz tool for Big Data Analytics with a novel interleaving mutation strategy that injects branch-related columns from existing tests.
- I work on evaluation against ML-based synthetic table generators like SDV, and conduct quantitative evaluation on test naturalness.
- Our tool is the first to consider naturalness in test generation without sacrificing code coverage and fault detection capability.

## **Purdue Security Lab**

Summer Undergraduate Research Intern  
Advisor: **Prof. Xiangyu Zhang**

Remote, Purdue University  
*Jul. 2021 - Sep. 2021*

- **Pelican: Exploiting Backdoors of Naturally Trained Deep Learning Models In Binary Code Analysis**

[5] Zhuo Zhang, Guanhong Tao, Guangyu Shen, Shengwei An, Qiuling Xu, Yingqi Liu, Yapeng Ye, **Yaoxuan Wu**, Xiangyu Zhang. Pelican: Exploiting Backdoors of Naturally Trained Deep Learning Models In Binary Code Analysis. *USENIX Security 2023*.

- A inversion trigger technique to confuse DL-based binary analysis models.
- I conduct preliminary experiment to inject binary patterns to mislead a function boundary detection model BiRNN-func.
- Our tool can induce misclassification on all 15 target binary analysis models.

## **Programming Languages Lab at Peking University**

Undergraduate Student Researcher  
Research Advisor: **Dr. Xin Zhang**

Peking University  
*Oct. 2020 - Jun. 2022*

- **Optimizing Probabilistic Inference in Bayesian Program Analysis**

- An accelerator for probability inference in Bayesian program analysis.
- I use Metropolis-Hastings-based stochastic optimization to find a smaller semantic-preserving derivation graph, achieving  $1.3\times$  reduction.
- I optimize existing message passing algorithm in probability inference, with a theoretical exponential boosting, achieving  $2\times$  empirical acceleration.

## **Collaboration**

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I have had the privilege of collaborating with faculty members, including Miryung Kim, Muhammad Ali Gulzar, Remy Wang, Xin Zhang, Guy Van den Broeck, and Xiangyu Zhang. I have also had the privilege to work with Brandon Paulsen and Joey Dodds during my internship.

## Awards

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**2024-** Amazon Fellowship

**2023-** UCLA Graduate Summer Research Scholarship

**2022-** UCLA Computer Science Department Scholar Award

**2018-2021** ACM-ICPC Asia Nanjing Gold Medal, EC Final Gold Medal, Beijing Silver Medal

**2021-** Jiukun Scholarship

**2020-** John Hopcroft Scholarship

**2019-** Peking University Third Prize Scholarship

**2018-** Peking University Freshman Scholarship

**2017-** National Olympics in Informatics (China) Gold Medal